LESSON 6.11

Fraction Division Problem Solving

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can solve real-world problems by deciding whether the question asks how many groups fit or how much is in one group, writing a fraction division equation, and checking the solution by multiplying.

WHY THIS MATTERS Agent Torres is closing the biggest case of the year. She has 2/3 of a gallon of fingerprint dust left and each test uses 1/6 of a gallon. Before the evidence expires, she needs to figure out exactly how many tests she can run. Can you write the equation and solve it?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Model *Spanish: Modelo*	A picture or math way to show a problem so you can solve it.	A bar model showing 3/4 split into 1/8-size pieces, or the equation 3/4 ÷ 1/8 = 6
Equation *Spanish: Ecuación*	A math sentence with an equal sign showing both sides are the same.	3/4 ÷ 1/8 = 6 — the left side (division) equals the right side (answer)
Solution *Spanish: Solución*	A number that makes the equation or inequality true.	In 3/4 ÷ 1/8 = 6, the solution is 6 portions
Reasonableness *Spanish: Razonabilidad*	Checking if your answer makes sense.	3 ÷ 1/4 = 12. Is 12 reasonable? Yes, because 1/4 is small so many pieces fit into 3.
Inverse operations *Spanish: Operaciones inversas*	Two math actions that undo each other, like × and ÷.	If 3/4 ÷ 1/8 = 6, then 6 × 1/8 = 6/8 = 3/4. Multiplication checks division.

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Agent Torres has 2/3 gallon of dust and each test uses 1/6 gallon. I ask: how many tests fit?

1. The total 2/3 is the dividend. The size of each test 1/6 is the divisor. My equation is 2/3 ÷ 1/6.
2. In sixths, 2/3 = 4/6, and 4 sixths ÷ 1 sixth = 4. Multiplying by the reciprocal gives the same answer: 2/3 × 6/1 = 12/3 = 4.
3. So she can run 4 tests. I check by multiplying back: 4 × 1/6 = 4/6 = 2/3. It matches the total, so the answer is reasonable.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

A ribbon is 3/4 yard and each bow uses 1/4 yard. The total 3/4 is the dividend; 1/4 is the divisor.

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The total amount is the _____ because it is being split into groups. The size of each group is the _____ because we are dividing by that amount.

Word bank Model | Equation | Solution | Reasonableness | Inverse operations | dividend | divisor | total | group | split | equation

WATCH OUT!

A common mistake

A common mistake in Fraction Division Problem Solving is reversing which fraction is the dividend and which is the divisor when setting up the equation from a word problem. For example: a pitcher holds 3/4 cup of punch, and each serving is 1/8 cup

LESSON 6.11

Fraction Division Problem Solving

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can solve real-world problems by deciding whether the question asks how many groups fit or how much is in one group, writing a fraction division equation, and checking the solution by multiplying.

WHY THIS MATTERS Agent Torres is closing the biggest case of the year. She has 2/3 of a gallon of fingerprint dust left and each test uses 1/6 of a gallon. Before the evidence expires, she needs to figure out exactly how many tests she can run. Can you write the equation and solve it?

PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Sort each word problem: should you DIVIDE or MULTIPLY to solve it?

Groups: Division Problem | Multiplication Problem

_____ How many 1/3-cup scoops fit into 2 cups? (2 ÷ 1/3)

_____ A ribbon is 3/4 yard. Each piece is 1/8 yard. How many pieces? (3/4 ÷ 1/8)

_____ You run 1/2 mile, 3 times. How far total? (1/2 × 3)

_____ A recipe uses 2/3 cup. You make half a recipe. How much? (2/3 × 1/2)

HINT Re-read each story: does it ask 'how many fit?' or 'how much is a part of the amount?'

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

A ribbon is 4/5 of a yard long. Each bow uses 1/10 of a yard. Which equation finds how many bows can be made?

(A) 1/10 ÷ 4/5
(B) 4/5 ÷ 1/10
(C) 4/5 × 1/10
(D) 4/5 + 1/10

HINT Name the roles: 4/5 yard is ALL the ribbon; 1/10 yard is what one bow uses.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

A detective has 1/2 gallon of solution. Each test uses 1/8 gallon. How many tests can be run?

(A) 1/16
(B) 1/4
(C) 4
(D) 16

HINT The total is 1/2 gallon; each test drains 1/8 gallon — you're counting tests.

Rewrite each fraction _____

Work it out _____

Simplify _____

1. **Same size pieces** *Common denominator first.*
2. **Operate** *Only then add or subtract.*
3. **Simplify** *Divide out common factors.*

4 **Circle the letter of the best answer. Show how you know.**

A trail is 3/4 mile long. A jogger runs laps of 1/4 mile each. How many laps does the jogger complete?

(A) 1/3 lap
(B) 3 laps
(C) 3/16 lap
(D) 4 laps

HINT The trail is 3/4 mile total, and every lap covers exactly 1/4 mile.

Rewrite each fraction _____

Work it out _____

Simplify _____

1. **Same size pieces** *Common denominator first.*
2. **Operate** *Only then add or subtract.*
3. **Simplify** *Divide out common factors.*

5 **Circle the letter of the best answer. Show how you know.**

A board is 5/6 foot long. Each shelf bracket needs 1/6 foot. How many brackets fit?

(A) 1/36
(B) 5
(C) 5/36
(D) 6

HINT Board = 5/6 foot, each bracket = 1/6 foot — both are already in sixths.

Write any whole number over 1 _____

Keep · Change · Flip _____

Multiply across _____

Simplify · label the units _____

1. **Whole over 1** *3 becomes 3/1.*
2. **Keep** *The first fraction does not move.*
3. **Change · Flip** *÷ becomes ×; flip the second fraction.*
4. **Check** *Dividing by a piece smaller than 1 gives a BIGGER answer.*

PART 3 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

Spot the Common Mistake

1. **Problem** A detective has 4/5 gallon of cleaning fluid. Each spray bottle holds 1/5 gallon. How many bottles can she fill?
2. **Student's equation** 1/5 ÷ 4/5
3. **Solve** 1/5 × 5/4 = 5/20 = 1/4
4. **Answer** 1/4 of a bottle

Which step has the mistake? Explain what went wrong, then show the correct work.

HINT Which value is the total being split up — 4/5 or 1/5?

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How do you decide which value is the dividend and which is the divisor in a word problem?

SENTENCE STARTER The total amount is the ___ because it is being split into groups. The size of each group is the ___ because we are dividing by that amount.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 6.11

Fraction Division Problem Solving

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can solve real-world problems by deciding whether the question asks how many groups fit or how much is in one group, writing a fraction division equation, and checking the solution by multiplying.

WHY THIS MATTERS Agent Torres is closing the biggest case of the year. She has 2/3 of a gallon of fingerprint dust left and each test uses 1/6 of a gallon. Before the evidence expires, she needs to figure out exactly how many tests she can run. Can you write the equation and solve it?

PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Solve each word problem by writing and solving an equation.

Problem	Equation	Solution
A board is 7/8 foot long. Each bracket needs 1/4 foot. How many FULL brackets can be cut?		
A jug holds 5/6 liter. Each cup holds 1/3 liter. How many FULL cups can be filled?		
A path is 3/4 mile long. A runner does laps of 3/8 mile. How many laps fit?		

SHOW YOUR WORK

2 **Write the name of the correct group on each line.**

Sort: is the equation and solution CORRECT, or is there an error?

Groups: Correct Equation and Solution | Incorrect Equation or Solution

_____ A tank has 5/6 gallon. Each bucket holds 1/3 gallon. → 5/6 ÷ 1/3 = 2 1/2 buckets

_____ A pipe is 2/3 foot long. Each piece is 1/6 foot. → 2/3 ÷ 1/6 = 4 pieces

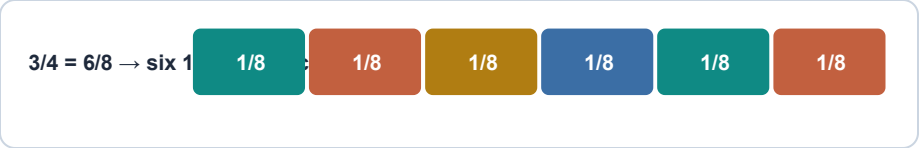
_____ A bag has 3/4 lb. Each scoop is 1/8 lb. → 3/4 ÷ 1/8 = 6 scoops

_____ A rope is 1/2 yard. Each section is 1/4 yard. → 1/4 ÷ 1/2 = 1/2 section

_____ A jug has 4/5 liter. Each cup is 2/5 liter. → 4/5 × 2/5 = 8/25 cups

_____ A shelf is 7/8 foot. Each book is 1/4 foot. → 7/8 ÷ 1/4 = 7/32 books

3 Use the model to solve. Show your work.



How many 1/8-pound molds can be made from 3/4 pound?

SHOW YOUR WORK

Answer: _____

PART 2 Apply it

real-world problems

4 Circle the letter of the best answer. Show how you know.

Marcus solved 2/3 ÷ 1/4 and got 8/3 = 2 2/3. He says 'That can't be right because I started with less than 1.' Is Marcus's math correct? Is his reasoning correct?

(A) His math and reasoning are both correct — the answer should be less than 1

(B) His math is wrong — the answer should be 1/6

(C) His math is correct (2 2/3), but his reasoning is wrong — dividing by a small fraction gives a larger result

(D) His math is wrong — the answer should be 8/12

SHOW YOUR WORK

5 Circle the letter of the best answer. Show how you know.

Agent Torres poured 3/4 gallon of fingerprint dust into a jar, and that filled 1/2 of the jar. How much dust fills the WHOLE jar? (3/4 ÷ 1/2 asks how much is in one whole jar.)

(A) 3/8 gallon

(B) 2/3 gallon

(C) 1 1/2 gallons

(D) 1 1/4 gallons

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 Find the mistake. Explain it, then show the correct work.

Spot the Common Mistake

1. **Problem** A chemist has 3/4 liter of solution. Each test tube holds 1/8 liter. How many test tubes can be filled?
2. **Student's equation** 3/4 ÷ 1/8 = 3/4 × 1/8
3. **Solve** 3/4 × 1/8 = 3/32
4. **Answer** 3/32 of a test tube

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How do you decide which value is the dividend and which is the divisor in a word problem?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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LESSON 6.11

Fraction Division Problem Solving

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Solve each multi-step problem. Show the equation and verify with multiplication.

Problem	Equation	Solution	Check (quotient × divisor = dividend)
A chef has 2 1/2 cups of flour. Each batch of cookies needs 1/4 cup. How many batches?			
A detective has 1 1/3 gallons of solution. Each test uses 2/3 gallon. How many tests?			
A board is 3 3/4 feet long. Each piece must be 5/8 foot. How many pieces?			

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

2 Find the mistake. Explain it, then show the correct work.

Find the Problem-Solving Error

1. **Problem** A tank has 3/4 gallon of water. Each bucket holds 1/6 gallon. How many buckets can be filled?
2. **Student's equation** 1/6 ÷ 3/4
3. **Solve** 1/6 × 4/3 = 4/18 = 2/9
4. **Answer** 2/9 buckets

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 Answer in complete sentences.

Write your own fraction division word problem where the answer is 8. Then show the equation, solve it step by step, and explain how you know your answer is reasonable.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

How do you decide which value is the dividend and which is the divisor in a word problem?

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